Technische Hochschule Augsburg

STATISTICS

International Information Systems, 3rd semester

(2) Descriptive statistics: measures of central tendency, frequency distributions

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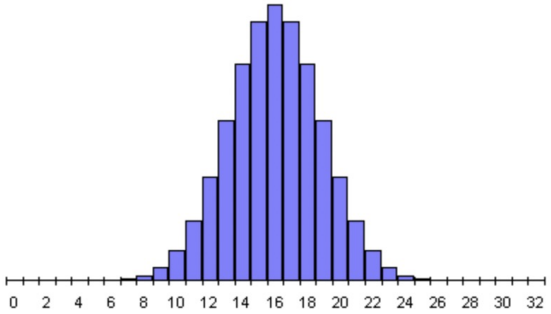
TODAY'S TOPICS AND LEARNING GOALS

Descriptive and Inferential Statistics
Introduction to Descriptive Statistics

- understand and explain the difference between ***sample*** and ***population***, ***descriptive*** and ***inferential*** statistics, ***parameters*** and ***statistics***

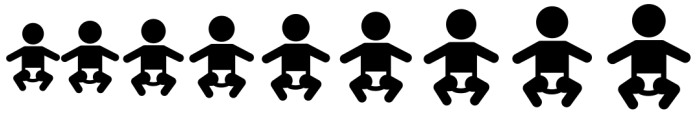
Frequency Distributions

- describe and visualize the ***shape*** of the distribution of your data and the properties of the distribution



Measures of Central Tendency

- compute the **mode, mean, median** and understand what they tell us about the data (and what they don't)



DESCRIPTIVE AND INFERENTIAL STATISTICS

DESCRIPTIVE STATISTICS AND INFERENTIAL STATISTICS

What's the distribution of Halloween candy in Germany?

population

What's the distribution of Halloween candy in the group I have collected?

sample

Inferential Statistics

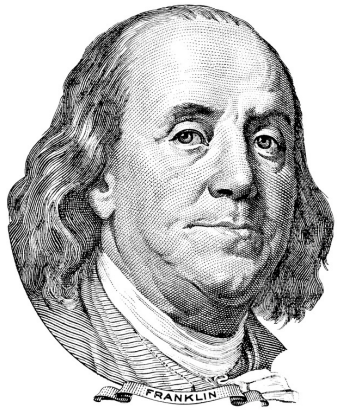
- using the sample to estimate the features of the population
- tests hypotheses ("there is significantly more gummy worms than smarties")

Descriptive Statistics

- a snapshot of the sample

TRUTH AND UNCERTAINTY

"In this world nothing can be said to be certain,
except death and taxes"
Benjamin Franklin



- We can never be 100% sure that our **conclusions about the population** are correct
  - Inferential statistics always brings ***some degree of uncertainty***
- Statistics is only a tool – We have to make decisions **ourselves**
  - based on this information & despite the uncertainty
  - some decisions need stronger evidence / less uncertainty than others!

There's no "magical AI / software" which will do this for you!

SAMPLE AND POPULATION

Why do we need a sample?

"In this quarter, the purchase of vegan candy has increased compared to last year"

=

population *sample*

The population is small enough that we can use it all as a sample and look at it *directly*

"In Germany, the purchase of vegan candy has increased compared to last year"

>

population *sample*

The population is too big to be measured directly: We need to extract a smaller sample and use it to estimate the features of the population

Inferential Statistics

How sure are we of the conclusions we draw about the population based on the sample?

SAMPLE AND POPULATION

Why do we need a sample?

“In this quarter, the purchase of vegan candy has increased compared to last year”

=

population *sample*

parameters

↗ the values of the population we are interested in

“In Germany, the purchase of vegan candy has increased compared to last year”

>

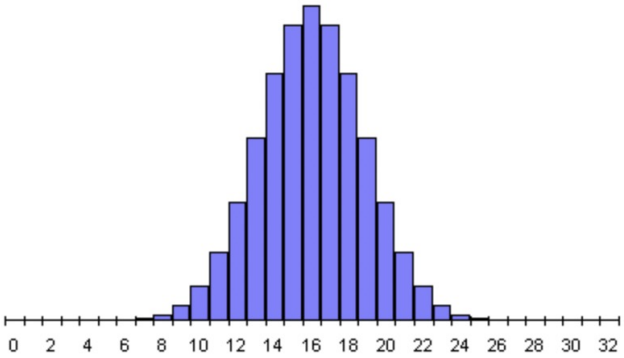
population *sample*

Inferential Statistics

statistics

↗ the values we can actually compute from the sample

INTRODUCTION TO DESCRIPTIVE STATISTICS



DESCRIPTIVE STATISTICS

A snapshot of your data

The purpose of **descriptive statistics** is to describe **a set of data** by

- ***organizing and simplifying*** the data
- reporting measures ***summarizing*** the data
- ***visualizing*** the data
- showing ***patterns***
- making the information from raw data ***manageable*** and easy to ***communicate***

DESCRIPTIVE STATISTICS

A snapshot of your data

We will look at:

- Frequency distributions
- Measures of central tendency

Today

- Measures of dispersion
- Outliers

Next week

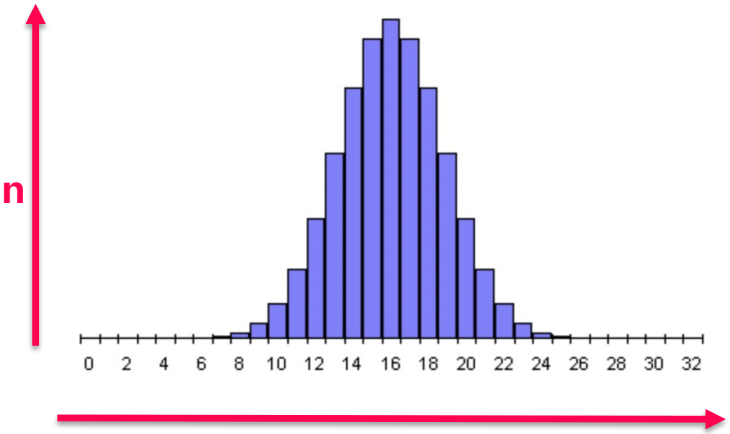
Questions?

FREQUENCY DISTRIBUTIONS

FREQUENCY DISTRIBUTIONS

Histograms

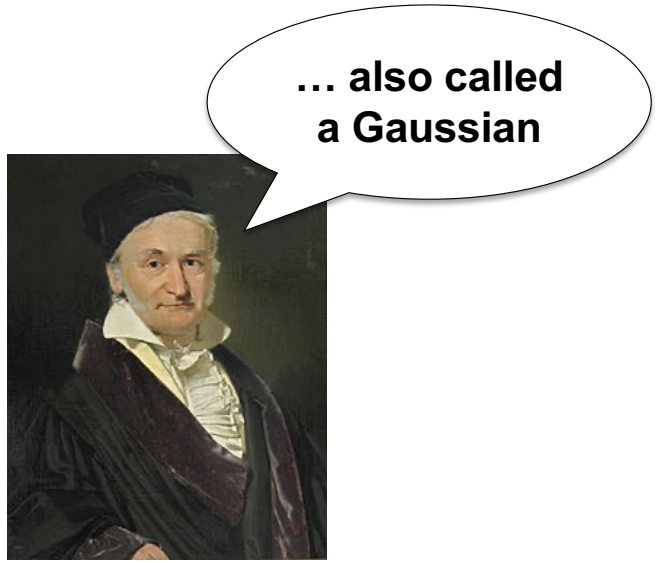
- Datasets typically contains hundreds or thousands of datapoints
  - It's not possible for us to interpret the data at a glance
  - Plotting data can reveal much more than just looking at raw data
- **Frequency distributions** or **histograms** of the data are a good first step to assess the properties of your data
  - they represents how many times a certain event or observation occurred
  - they come in many different shapes and sizes



FREQUENCY DISTRIBUTIONS

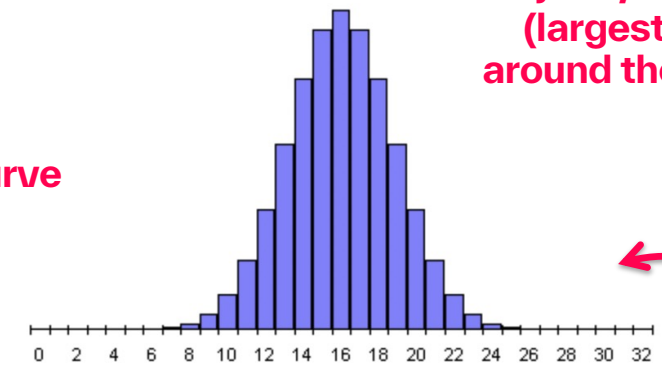
Normal distributions

- Data in a **normal distribution** is distributed ***symmetrically*** around the center
- A lot of naturally-occurring phenomena are normally distributed
  - Height, blood pressure, precipitation, sunshine hours in a year



majority of scores
(largest bars)
around the *center*

bell-shaped curve



deviating scores are less frequent in the
tails (smaller bars)

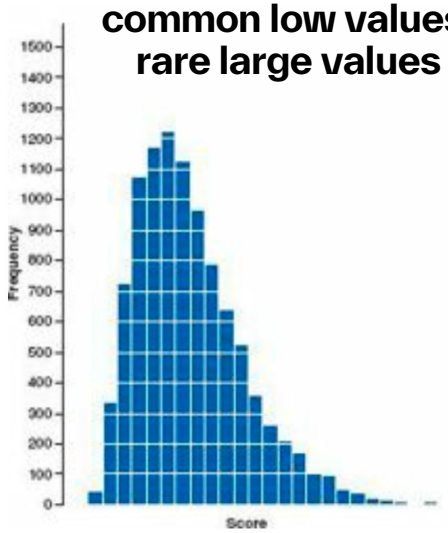
FREQUENCY DISTRIBUTIONS

Skewed distributions

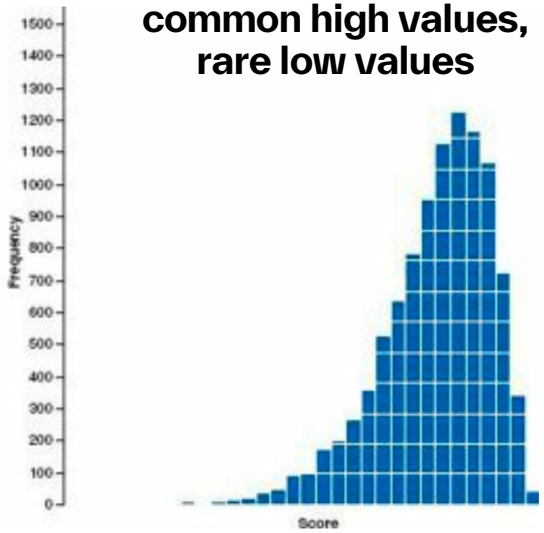
- Data in a **skewed distribution** are distributed ***at one end of the scale*** (**skew** = lack of symmetry)
  - Reaction times in psychology studies, average income, number children per family are positively skewed
  - Age of death, retirement age, gestational age at delivery (live births) are negatively skewed

positively skewed or right-skewed distribution: majority of scores to the lower end of the scale

lower bound, plausible / common low values, rare large values



higher bound, plausible / common high values, rare low values



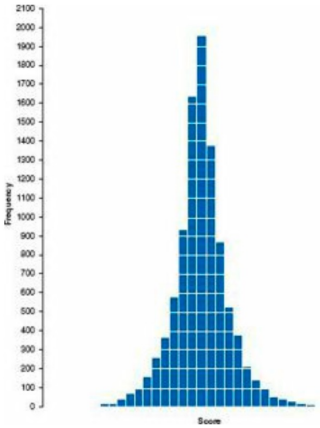
negatively skewed or left-skewed distribution: majority of scores to the higher end of the scale

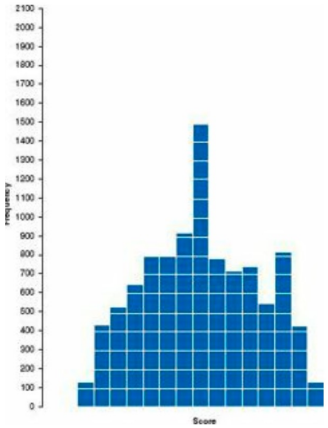
FREQUENCY DISTRIBUTIONS

Kurtosis

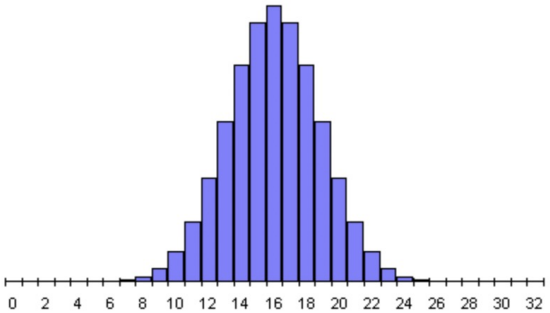
- The kurtosis of a distribution refers to the degree to which scores cluster at the tails

positive kurtosis (heavy-tailed or leptokurtic distribution): many scores in the tails, “pointy”





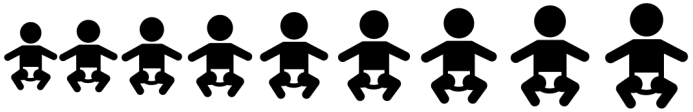
negative kurtosis (light-tailed or platykurtic distribution): few scores in the tails, “flat”



- Normal distributions are not skewed (skew = 0)
- Normal distributions are not flat or pointy (kurtosis = 0)
- A skew or a kurtosis above and below 0 is a deviation from the normal distribution

Questions?

MEASURES OF CENTRAL TENDENCY



MEASURES OF CENTRAL TENDENCY

- A way to capture a snapshot of your data (an “id” of your data) is to find its **central tendency** (the center of the frequency distribution)
- Measures of central tendency:
  - the ***mean***
  - the ***median***
  - the ***mode***

MEASURES OF CENTRAL TENDENCY

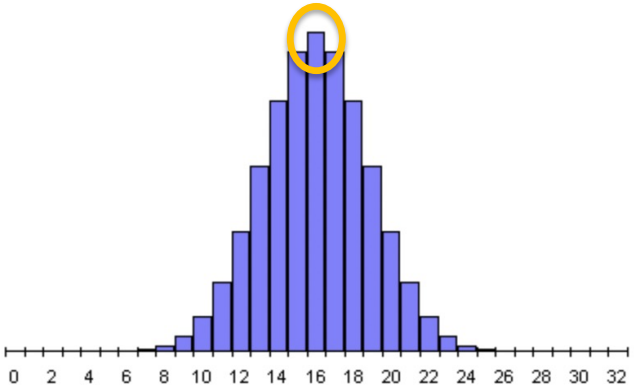
The mode

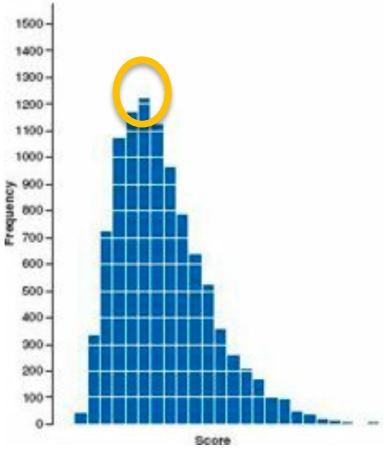
- The **mode** is simply the score that occurs ***most frequently*** in the data set

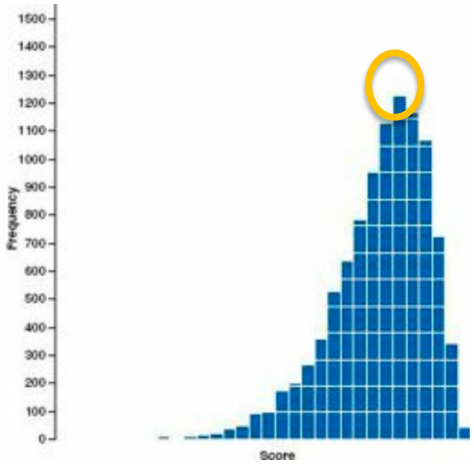
  1, 1, 0, 2, 1, 0, 3, 2, 2, 2, 4, 2, 1, 3, 3

- How do you find the mode in a frequency distribution?
  - it's the tallest bar! (regardless of the other bars)
  - it's a score that ***actually occurred*** in the data and the ***most likely*** to be drawn at random
  - if ***two adjacent scores*** are the most (equally) frequent, typically you take **their average** as the mode
- useful for **nominal data (frequency matters)**

Value	Frequency
0	2
1	4
2	5
3	3
4	1



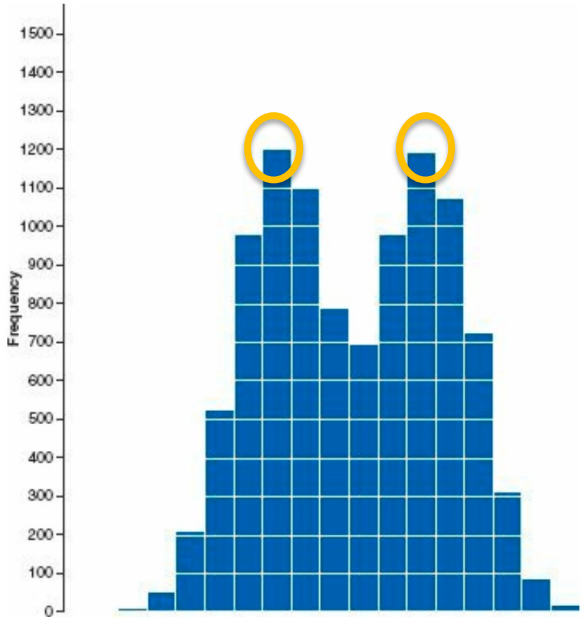




MULTIMODAL DISTRIBUTIONS

- What if there are **more modes**, that is *non-adiacent* scores with equal (or nearly equal) high frequency?
  - a distribution with more than one peak is called **multimodal distribution**
  - **bimodal** if there are two peaks,
    unimodal if there's only one peak

- A multimodal distribution may indicate that your sample has several **patterns of response**
  - Maybe it's actually two distributions (e.g. scores of students who studied, scores of students who did not)



MEASURES OF CENTRAL TENDENCY

The median

- The **median** is the ***middle score*** in the data set when scores are ranked / ordered

1, 1, 0, 2, 1, 0, 3, 2, 2, 2, 4, 2, 1, 3, 3
0, 0, 1, 1, 1, 1, 2, 2, 2, 2, 2, 3, 3, 3, 4

- If the number of score is even (there is no “middle score”),
  typically, you take the average of the two middle scores as the mode
- The median is relatively unaffected by skewed distributions and extreme scores
  - e.g. salaries, home prices
- It can’t be used with nominal data, but works for ordinal, interval and ratio scale data

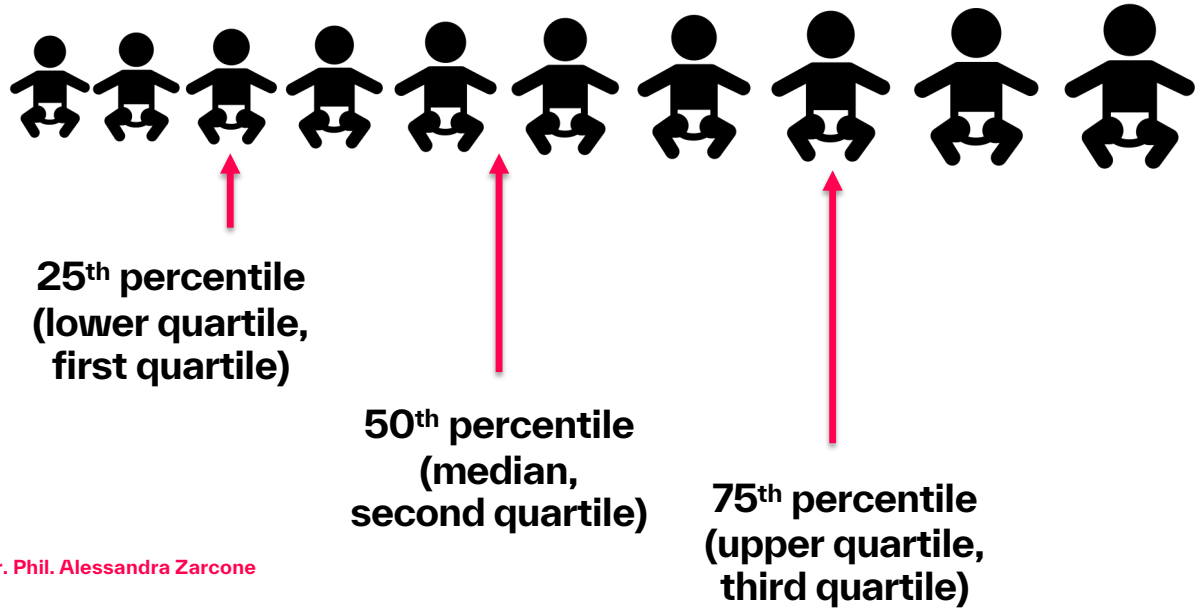
MEASURES OF CENTRAL TENDENCY

Percentiles and Quartiles

- The median is also called the **50th percentile**
  - **xth percentile** = the point at or below which x% of the scores fall

quartiles
divide the data points into four quarters:

the 1st quartile has 1/4 of the data before it, the 2nd quartile 2/4 , etc.



MEASURES OF CENTRAL TENDENCY

The mean

- The **mean** (**average**) is the sum of all the scores divided by the number of scores

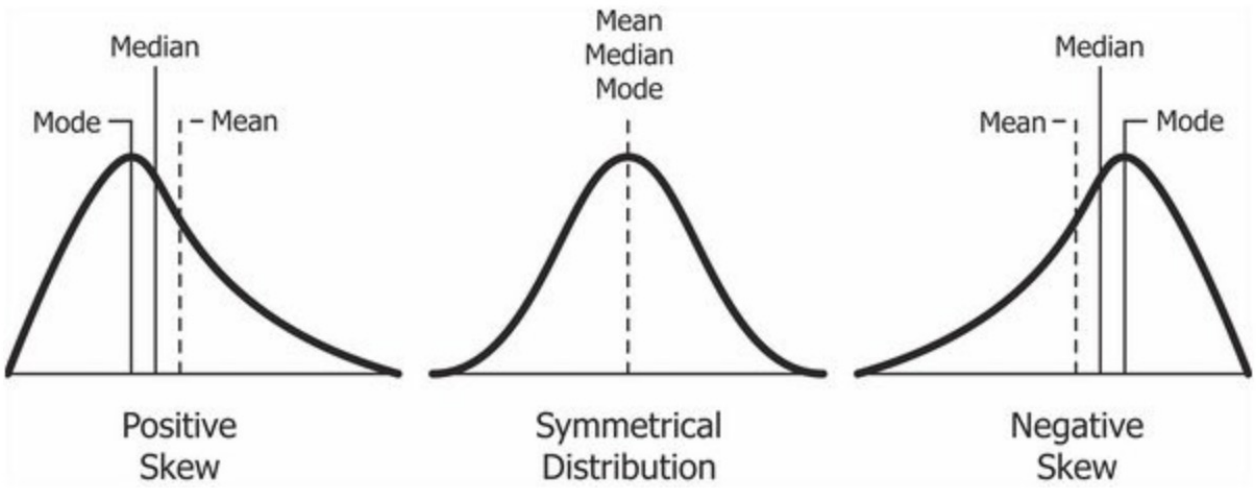
$$\bar{X} = \frac{\sum_{i=1}^{n} x_i}{n}$$

- The mean is the most common measure of central tendency
- It can be manipulated algebraically
- It is a better estimate of the population's tendency than the median or the mode

MEASURES OF CENTRAL TENDENCY

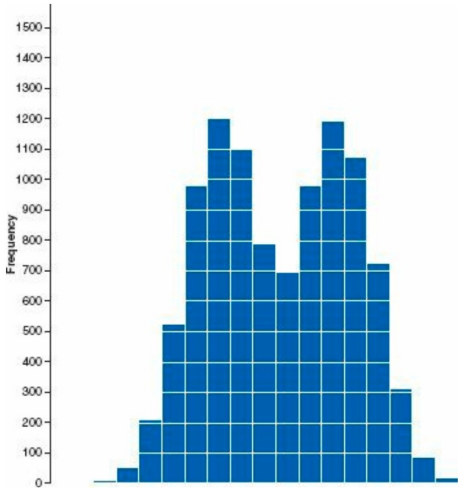
The mean

- The mean is influenced by extreme scores
  - **symmetric and unimodal distributions**: mode, mean and median are in agreement
  - **skewed distributions**: mean, median, and mode can all be quite different



Questions?

SUMMING UP



- We usually do not look at the whole phenomenon (**population**) but at a smaller, observable **sample**
  - **Descriptive statistics** provides a summary of the observations about the **sample**
  - **Inferential statistics** uses the sample statistics to estimate the parameters of the **population** to a certain **degree of certainty**
- The purpose of **descriptive statistics** is to summarize and visualize the data and understand patterns
- A good first step is to **plot the frequency of your data** using a **histogram**
  - Frequency distributions of data come in **different shapes** depending on their symmetry / skew, their kurtosis, the number of peaks (“modes”)
- **Mean, median and mode** are typical **measures of central tendency**
  - They differ with regard to the scales they can be used on, their existence in the dataset, their sensitivity to extreme values, their versatility

HOMEWORK

- Chapter 4, sections 4.1 to 4.4 of Howell,
  "Fundamental Statistics for the Behavioral Sciences" (available on Moodle)

- Watch this video:
  - "Mean, Median, and Mode: Measures of Central Tendency" (Crash Course Statistics)
    https://www.youtube.com/watch?v=kn83BA7cRNM

Questions?